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Cystic Fibrosis

Kimberly M. Dickinson, MD, MPH¹, Joseph M. Collaco, MD, PhD¹

¹Johns Hopkins University, School of Medicine, Eudowood Division of Pediatric Respiratory Sciences, Baltimore, MD

Abstract

Cystic fibrosis (CF) is one of the most commonly diagnosed genetic disorders. Clinical characteristics include progressive obstructive lung disease, sinusitis, exocrine pancreatic insufficiency leading to malabsorption and malnutrition, liver and pancreatic dysfunction, and male infertility. Although CF is a life-shortening disease, survival has continued to improve to a median age of 44.4 years due to earlier diagnosis through routine newborn screening, promulgation of evidence-based guidelines to optimize nutritional and pulmonary health, and development of CF-specific interdisciplinary care centers. Future improvements in health and quality of life for individuals with cystic fibrosis are likely with the recent development of mutation-specific modulator therapies. In this review, we will cover the current understanding of the disease manifestations, diagnosis, and management as well as common complications seen in individuals with CF.

Introduction

Cystic fibrosis (CF) is a common life-shortening autosomal recessive genetic disorder, characterized by pulmonary manifestations, specifically chronic and progressive obstructive lung disease, sinusitis, malabsorption due to pancreatic exocrine insufficiency leading to malnutrition, liver disease (biliary cirrhosis), and CF-related diabetes mellitus (CFRD). Earlier diagnosis through newborn screening, improved therapies to optimize lung health and nutritional status, as well as aggressive treatment of chronic respiratory infections and lung transplantation for end-stage lung disease have led to significant improvements in survival. CF was a uniformly fatal disease in childhood at the time of its initial description in 1938, (1) but the predicted median survival is currently 44.4 years. Currently, over 50% of people living with CF are 18 years and older, (2) resulting in the evolution of the disease from exclusively a disease of childhood to a disease where affected individuals transition to adulthood and adult providers. With the advent of newer therapies targeting the basic genetic defect that causes the disease and the expansion of the age and genetic variants for which these therapies are indicated, there is promise of continued improvement of quality of life as well as overall health and survival. Individuals with CF benefit from coordination of care between their primary care providers and their interdisciplinary CF care team, in

Address correspondence to: Joseph M. Collaco, Eudowood Division of Pediatric Respiratory Sciences, Johns Hopkins University, 200 N. Wolfe Street, Room 3071 Baltimore MD, 21287, [mcollac1@jhmi.edu], 410-955-2035.

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addition to routine visits with both. This review will cover the current understanding of the disease manifestations, diagnosis, and management as well as common complications seen in individuals with CF.

Epidemiology

CF is one of the most common genetic disorders among Caucasians, with an incidence of 1:3,200 individuals. (3) The incidence does vary significantly by race/ethnicity with an incidence of 1:13,500 in people of Hispanic background, 1:15,000 in people of African descent and 1:35,000 in people of Asian descent. (4) It is estimated that one in every 35 Americans is a carrier of CF. (5) Based on statistics from the 2018 Annual Data Report published by the Cystic Fibrosis Foundation (CFF), there are an estimated 30,000 affected individuals in the United States living with CF. (2) Worldwide, there are an estimated 70,000 people, with the highest prevalence in North America, Europe and Australia. Annually, approximately 1,000 new cases are diagnosed in the United States (although the rate of new cases may be declining related to pre-conception screening), (6) and the incidence is equal in males and females. Since 2010, all 50 states screen neonates for cystic fibrosis, (7) and as a result 60% of new diagnoses occur via newborn screening. (2)

Pathogenesis

CF is a multisystem disorder that results from deleterious genetic variants in the *CFTR* gene located on the chromosome 7q31.2, which encodes for the cystic fibrosis transmembrane conductance regulator (CFTR) protein. Defects in this protein lead to absent or malfunctioning chloride channels in the apical membranes of the lung surface or glandular epithelium resulting in the formation of thick and sticky mucus, leading to chronic lung infections, pancreatic and liver dysfunction, and reduced fertility. CF also results in abnormal chloride channel function in the sweat glands, resulting in excessive salt loss in sweat, an observation first studied by Dr. Paul di Sant'Agnese after caring for infants with CF presenting with dehydration during a heat wave in New York City in 1948. (8) This clinical observation paved the way for the pilocarpine iontophoresis sweat test for CF diagnosis. (9)

CF is an autosomal recessive disorder and for individuals to have CF, they must inherit 2 deleterious *CFTR* variants. To date, there are more than 2,000 different *CFTR* variants reported, some of which are confirmed to cause CF, and others with more putative links to the disease. (10)(11) They are classified into six distinct groups that reflect abnormalities of CFTR protein synthesis, structure and function (Figure 1). The most common CF-causing variant is *F508del* (p.Phe508del). *F508del* is a Class II mutation, meaning that the CFTR protein is created, but misfolding prevents it from reaching the cell surface ("trafficking defect"). Overall, 44.2% of individuals with CF are homozygous for *F508del*, an additional 40.5% have one copy of *F508del* and another variant, and 15.3% have 2 non-*F508del* variants. (2) The specific *CFTR* variants that an individual carries determines the amount of functioning CFTR protein present, and is partially correlated with phenotypic severity and organ involvement. (12)

Clinical Presentations

The classic manifestations of CF include a triad of (1) recurrent sinus and pulmonary infections, (2) steatorrhea, and (3) malnutrition, which in its most severe form presents as failure to thrive. In the lungs, mucous plugging from dehydrated thick secretions results in inflammation, chronic infection, progressive small airways obstruction and the development of bronchiectasis, which is an abnormal, permanent enlargement of the bronchi (Figure 2). Bronchiectasis leads to decreased ability to clear secretions, causing increased rates of infections, which further dilates and damages the airways. In addition, the effects of diminished or absent chloride channel function can result in dysfunction in several other organ systems (Figure 3). Pancreatic involvement includes both pancreatic exocrine insufficiency, which results in fat, protein and carbohydrate malabsorption and subsequent malnutrition, as well as insulin insufficiency and the development of CFRD.

In infants and young children, other presentations may also be indicative of CF. *In utero*, ultrasonographic evidence of hyperechogenic or dilated bowel are suggestive of intestinal obstruction, which have been reported in 50% to 78% of fetuses affected with CF. (13)(14) Postnatally, delayed meconium passage or meconium ileus is present in 11.9% of infants <1 year of age with CF (2), and results from thick GI secretions that become adherent to the intestinal mucosa leading to bowel obstruction. Meconium ileus is often accompanied by abdominal distension and dilated loops of bowel on imaging and a reported 30% cases of meconium ileus are complicated by intestinal perforation and peritonitis. (2) Approximately 20% of untreated children (6 months to 3 years) have rectal prolapse, which is secondary to malabsorption, malnutrition, and bulky stools as opposed to constipation. (15) Other clinical presentations during the neonatal period may include prolonged jaundice secondary to biliary stasis or bile duct obstruction and hemorrhagic disease of the newborn owing to vitamin K deficiency. Throughout infancy and early childhood, individuals may also present with salt depletion syndrome characterized by a hyponatremic, hypochloremic, hypokalemic metabolic alkalosis, and edema/acrodermatitis due to hypoproteinemia from malabsorption. (16)

Typical respiratory findings in older children, adolescents, and adults who newly present with CF may include recurrent sinusitis, bronchitis, or pneumonias, asthma that is poorly responsive to standard management, nasal polyposis or digital clubbing on physical exam, as well as bronchiectasis on lung imaging studies. (16) Common gastrointestinal symptoms may include malnutrition, poor growth, steatorrhea, intestinal obstruction, chronic constipation, rectal prolapse, and liver disease. (17) Individuals with pancreatic sufficient CF (who are more likely to be diagnosed later in life due to appropriate weight gain) can present with pancreatitis secondary to progressive pancreatic inflammation, although the exact etiology is unclear. (18) Finally, over 98% of men with CF are infertile as a result of obstructive azoospermia secondary to congenital bilateral absence of the vas deferens and may be diagnosed during an infertility work-up. (19) Newborn screening is expected to reduce, but not eliminate late clinical presentations given the expected rate of false negatives associated with screening. Clinicians should include cystic fibrosis in the differential for unexplained recurrent respiratory bacterial infections (pneumonia, bronchitis, persistent cough, and/or sinusitis) and/or failure to thrive.

SCREENING

Newborn screening (NBS) for CF was first introduced in the 1980s, and is performed by measuring an immunoreactive trypsinogen (IRT) level in screening neonatal blood spots. In neonates with CF, mucus plugs partially block the pancreatic ducts that lead into the small intestine, preventing trypsinogen from reaching the intestine and being converted to the pancreatic enzyme trypsin, even in infants with CF who are pancreatic sufficient. If an infant has an elevated IRT level, almost all U.S. state laboratories perform confirmatory *CFTR* variant testing (IRT/DNA) with one state laboratory currently repeating the IRT measurement (IRT/IRT). Screening is considered positive if the IRT level remains persistently elevated between the ages of 7 and 14 days of life, or if at least one deleterious *CFTR* variant is identified on genetic testing. A positive NBS result will trigger notification of either a neonatal intensive care provider or primary care provider, and the infant should be referred to a CFF-accredited center for definitive evaluation and sweat testing within 72 hours of a positive screening result. (20) Care centers may be located on the CFF website (www.cff.org/ccd/).

The benefits of diagnosing asymptomatic infants with CF includes increased attention to early lung health to slow lung disease progression, optimization of nutritional status with early enzyme replacement and aggressive nutritional counseling, as well as providing psychosocial support to families to help prevent or delay serious complications. (16) Possible risks of NBS include an increased number of medical interventions, exposure to respiratory pathogens through attending CF clinic, financial hardships given the cost of CF-related therapies, possible iatrogenic complications (e.g., early exposure to therapies with side effects), and caregiver anxiety or guilt stemming from false-positive screening results due to perinatal asphyxia or other perinatal problems. (16)(21) False positive rates may also be higher in African-American children as they have higher IRT levels than Caucasians (22), but have a much lower risk of CF. NBS can also be falsely negative, particularly in neonates with meconium ileus (23) or those screened via IRT/DNA, as this testing may be less sensitive for picking up mutations in minority populations. (22) With current NBS practices, the possibility exists for a positive NBS and the identification of *CFTR* variants that do not meet clinical criteria for CF diagnosis in individuals with normal or intermediate sweat chloride testing, a syndrome known as *CFTR*-Related Metabolic Syndrome (CRMS)/Cystic Fibrosis Screen Positive, Inconclusive Diagnosis (CFSPID). (24)

DIAGNOSIS

The CFF published consensus guidelines in 2017 establishing that a diagnosis of CF can be made if an individual has a clinical presentation consistent with the disease i.e. (1) a positive newborn screening; (2) clinical features consistent with CF (the presence of characteristic phenotypes such as of chronic, recurrent sinus and pulmonary disease, nutritional and gastrointestinal abnormalities, urogenital abnormalities in males (e.g., absence of the vas deferens), and/or salt depletion syndromes, or (3) a positive family history of CF *and* evidence of *CFTR* dysfunction (e.g., sweat chloride concentration ≥ 60 mmol/L). (24)(25)

While prenatal screening and NBS have allowed for earlier detection of CF in asymptomatic individuals, the quantitative pilocarpine iontophoresis sweat test remains the gold standard

for the diagnosis of CF. The sweat test, developed by Lewis Gibson and Robert Cooke in 1959, (9) specifically measures the amount of chloride in a person's sweat. Sweat chloride testing should be performed as soon as possible after a positive NBS result. It can be performed as early as 48 hours after birth (as sweat sodium levels are transiently elevated in the first 24 hours), (26) but should be undertaken as soon as possible after 10 days of age and ideally by 4 weeks of age. Infants should weigh more than 2 kg or be corrected to 36 weeks gestation to increase the likelihood of adequate sweat collection. (24) Infants with meconium ileus and any infant or child with symptoms suggestive of cystic fibrosis such as recurrent bacterial respiratory infections and/or failure to thrive should receive sweat chloride testing regardless of age or NBS results. Any abnormal sweat test result should be repeated on a separate date, or confirmed with genetic testing.

Results from sweat chloride testing can be categorized into diagnostic, intermediate, and unlikely. Diagnostic sweat chloride values are ≥ 60 mmol/L, and require a confirmatory second sweat test or 2 identified CF-causing genetic variants to make the diagnosis. Intermediate values are between 30-59 mmol/L, and sweat chloride testing should be repeated periodically in these individuals and further evaluation at a CF center should be considered. A diagnosis of CF can still be made in an individual with an intermediate value if the individual has 2 identified CF-causing genetic variants. Individuals with an intermediate sweat chloride value of 30-59 mmol/L and 0-1 CF-causing genetic variants may be diagnosed with a CFTR-related disorder depending on clinical presentation and family history. Individuals with sweat chloride values < 30 are unlikely to have CF, but if 2 CF-causing genetic variants are identified, they should still be diagnosed with CF. (24)

In conjunction with sweat testing, genetic testing is now widely available to help confirm a diagnosis of CF, particularly for cases with intermediate sweat chloride values. Identifying an individual's specific CF-causing variants is needed for prescribing CFTR modulator therapies, which are approved for particular *CFTR* variants. A diagnosis by genotype can be made with the identification of 2 known pathogenic variants on separate chromosomes. Most individuals with CF can be diagnosed through commercial laboratories, which test for the most common *CFTR* variants, (27) but complete sequencing of the *CFTR* gene may be necessary to help confirm diagnosis in individuals with clinically atypical presentations. (16) Clinical information related to specific *CFTR* variants can be found online (www.cftr2.org).

The era of genetic testing has expanded our understanding of CFTR dysfunction, but has also added complexity to the diagnosis of CF, as there are individuals with CF phenotypes without known CF-causing mutations, as well as individuals with detected mutations who remain asymptomatic. The limitations of sweat chloride testing and genetic testing may require performing both tests in selected patients where there is strong clinical suspicion of CF. (28)

CF Management

In the United States, individuals with CF should be evaluated at a minimum of quarterly in a CFF-accredited care center, (29) which can provide interdisciplinary, patient/family-centered care. Infants under 6 months of age should be evaluated monthly and then every 1-2 months

in the second six months of life (20) These regular visits allow for education surrounding airway clearance methods, infection prevention, monitoring of age-appropriate growth and weight gain, and in older children, the assessment of lung function.

Therapies to Maintain Optimal Lung Health

CF is characterized by viscous airway secretions that lead to chronic mucous obstruction, inflammation, and recurrent infections that result in long term damage to the airways (bronchiectasis) and lung parenchyma. (16) Chronic cough and sputum production are characteristic symptoms. Management of respiratory symptoms focuses on maintaining lung function and preventing the development of bronchiectasis and parenchymal destruction. (30) In addition to encouraging a smoke-free environment, children with CF are often prescribed several therapies, which will likely be lifelong and should be initiated shortly after diagnosis.

Clearance of Airway Secretions

A critical aspect of maintaining lung health is airway clearance therapy (ACT). By removing airway mucus, ACT helps to decrease the respiratory bacterial load along with irritants, leading to improved gas exchange and a decrease in airway obstruction. (16) Twice daily ACT as maintenance is recommended for all patients with CF, regardless of symptoms or disease severity, and is increased in frequency during acute CF pulmonary exacerbations (PEX). (31) Commonly used ACT modalities include manual percussion, positive expiratory pressure devices, and high-frequency chest wall oscillation (achieved through an inflatable vest that performs chest physical therapy by vibrating at a high frequency). No form of ACT has been demonstrated to be superior to any other form, and ACT choice should be personalized. Exercise should be encouraged as an adjunctive therapy, but should not be used as a substitute for airway clearance. (31) Nebulized agents that thin the viscous mucus of CF are commonly used with ACT, and include recombinant human rhDNase (dornase alfa/Pulmozyme) and hypertonic saline.

Chronic Airway Infections

In addition to patient education and infection control measures, aggressive management of chronic airway infections has been shown to prevent lung function decline. (32) Management includes frequent respiratory cultures (oropharyngeal or sputum), including surveilling for *Staphylococcus aureus* (particularly methicillin-resistant *S. aureus*) and *Pseudomonas aeruginosa*. (16) Receiving microbiology laboratories should be made aware of the patient's CF diagnosis in order to assess for the respiratory pathogens commonly seen in CF. The initial acquisition of *P. aeruginosa* is typically treated with anti-pseudomonal antibiotics such as nebulized tobramycin (TOBI nebs) in an attempt to achieve eradication. (33) Nebulized antibiotics such as tobramycin or aztreonam (Cayston) can also be used as suppressive therapy for individuals with chronic infection or colonization with *P. aeruginosa* and/or other gram-negative organisms. This suppressive therapy is administered every other month to decrease the risk of antibiotic resistance. Other organisms, including *Burkholderia cepacia* complex, nontuberculous mycobacteria (*Mycobacterium avium* complex and *Mycobacterium abscessus*) and fungal pathogens

(*Aspergillus fumigatus*) are also monitored, as they can have significant impact on CF lung disease. Of note, individuals with CF are also prone to developing a hypersensitivity reaction to *Aspergillus*, known as allergic bronchopulmonary aspergillosis or ABPA, which can affect lung function and requires management with corticosteroid therapy. (34)

Chronic Airway Inflammation

Cystic fibrosis lung disease is caused by a combination of infection and inflammation. The routine use of oral or inhaled corticosteroids in CF is not indicated unless used for another inflammatory comorbidity such as asthma or ABPA. (30) Chronic airway inflammation is managed with either high-dose ibuprofen or azithromycin. (35) Although ibuprofen has proven benefits in CF, the risk of gastrointestinal bleeding and need for monitoring serum levels has limited its use. (36) Azithromycin has been demonstrated to result in improved lung function and a reduced number of PEx, and is typically given orally three times per week. (37) However, there is concern that individuals with unrecognized mycobacterial infections receiving azithromycin long-term may develop resistance; screening using mycobacterial cultures is recommended before initiating treatment. (30)

Therapies to Maintain Optimal Nutritional Status

Decline in pulmonary status is the hallmark of CF, however poor growth is one of its earliest manifestations. The combined effects of decreased intake, malabsorption and increased metabolic demands contribute to the poor growth seen as early as infancy. Malnutrition has been associated with increased morbidity and mortality in CF. (38) CFF guidelines recommend that all children achieve a weight-for-length at or above the 50th percentile by 2 years old, and that all children and adolescents aged 2-20 years maintain a body mass index at or above the 50th percentile. (20)(39) Education about the role of enteral tube feeding in optimizing nutritional status should be provided to caregivers and patients throughout their lifetime. (40) Infants with CF should receive breastmilk if possible, and otherwise should receive standard infant formula (rather than hydrolyzed protein formulas). (20) Infants under 2 years old should be supplemented with table salt, up to ¼ teaspoon per day by 6 months of age, (20) due to ongoing salt losses. Infants with CF should have their fecal elastase measured after diagnosis to assess pancreatic functional status, as 85% of individuals with CF are exocrine pancreatic insufficient (fecal elastase <200 ug/g). (2) Pancreatic insufficiency leads to malabsorption, which presents as bulky, malodorous stools, malnutrition, and ultimately failure to thrive.

Pancreatic Enzyme Replacement Therapy

Pancreatic enzyme replacement therapy (PERT) should be initiated in those with a diagnostic fecal elastase or two *CFTR* variants associated with pancreatic insufficiency as well as those with unequivocal signs or symptoms of malabsorption. Individuals who have exocrine pancreatic insufficiency require enzyme replacement with every meal, snack, and enteral tube feeding, ranging from 2,000 to 2,500 units/kg of lipase per meal, to a maximum of 10,000 units/kg/day. (29)(41) Immobilized lipase cartridges may be used for patients on continuous enteral tube feeding to help hydrolyze fats in enteral formulas. Exceeding recommended dosages generally does not result in improved

nutrient absorption, and supratherapeutic dosing is associated with fibrosing colonopathy, an uncommon complication characterized by foreshortening and strictures of the right colon. (42) Individuals with CF can achieve age-appropriate growth with optimized PERT. (43)

Fat-Soluble Vitamin Replacement Therapy

Pancreatic insufficiency results in malabsorption of fat and associated fat-soluble vitamins A, D, E, and K. Vitamin A deficiency can be associated with night blindness and ocular xerosis, as well as dermatological manifestations, such as follicular hyperkeratosis. (16) Vitamin D deficiency may result in rickets, osteopenia, and osteoporosis, which can result in fractures; recommendations for individuals with CF specify maintaining a goal of a serum 25-hydroxyvitamin D level of at least 30 ng/ml. (44) Vitamin E deficiency may result in peripheral neuropathy, myopathy, and hemolysis while Vitamin K deficiency is associated with coagulopathy and may also contribute to bone disease. (16) Supplemental vitamin therapy should begin after diagnosis and annual monitoring of serum Vitamin A, D, and E levels by CF providers is recommended. (20)

CFTR Modulator Therapies

CFTR modulators are the first therapies to target the basic defect in CF by directly acting on the CFTR protein. They are categorized into 3 types: potentiators, correctors, and amplifiers. (45) Ivacaftor (Kalydeco), the first approved modulator therapy, is a potentiator, which helps improve chloride flow through the CFTR protein at the cell surface for patients with Class III-V mutations. Correctors, such as lumacaftor and tezacaftor, help the CFTR protein to form correctly to allow the protein to move, or traffic, to the cell surface. When added to potentiators, correctors such as lumacaftor/ivacaftor (Orkambi) or tezacaftor/ivacaftor (Symdeko) work to improve the amount of protein that reaches the cell surface for patients with Class II mutations. Kalydeco is currently Federal Drug Administration (FDA)-approved for individuals aged ≥ 6 months, Orkambi for those aged ≥ 2 years with homozygous *F508del* variants, and Symdeko for those aged ≥ 6 years with homozygous *F508del* or several other specific CF-variants.

In 2019, the FDA approved the use of a triple-combination therapy, elexacaftor/tezacaftor/ivacaftor (Trikafta), for individuals 12 and older with at least one *F508del* variant. This new therapy will allow nearly 90% of individuals with CF to have a highly effective therapy for the underlying cause of their disease. Trikafta has been shown in clinical trials to show dramatic improvement in key measures of disease, including increased lung function, reduced PEx, decreased sweat chloride values, increased BMI and improved patient-reported quality of life. Amplifiers are expected to increase the amount of CFTR protein that a cell makes, but are under development and not currently available clinically. (46)

Diagnosis and Management of Common Pulmonary and Extrapulmonary Complications of CF

Pulmonary exacerbation

One of the most common complications of CF lung disease is episodic acute worsening of symptoms, referred to as pulmonary exacerbations (PEx). PEx are characterized by increased respiratory symptoms, including coughing, sputum production, and/or wheezing, a decline in pulmonary function measures (specifically FEV₁), fatigue, decreased appetite, and weight loss. (47) Fevers are not commonly seen with PEx. (16) The frequency of PEx varies among individuals, but contributes to the long-term lung function decline of most people with CF. Treatment typically includes antibiotics and increased frequency of ACT to help clear secretions from the airways. Antibiotic therapy and mode of delivery (enteral, inhaled, and/or intravenous) is dictated by the severity of the exacerbation and previous/current respiratory culture results (Figure 4).

Hemoptysis

Hemoptysis is reported to occur in 3% of individuals with CF annually. (2) While often seen with severe lung disease, it can also be a manifestation of a PEx. (48) CF-related hemoptysis is most commonly a result of chronic infection and inflammation, leading to erosion of hypertrophied bronchial arteries into the airways. (49) Vitamin K deficiency, either from malabsorption or liver disease, can contribute to hemoptysis. Managing scant to moderate hemoptysis (≤ 240 mL) includes evaluation, likely antibiotic therapy for PEx management, and consideration of limiting certain exacerbating therapies, such as ibuprofen, hypertonic saline, DNase, and ACT. (48) Massive hemoptysis (> 240 mL) is considered life-threatening and management includes appropriate stabilization and discontinuation of anti-inflammatory and airway clearance measures. The treatment for massive hemoptysis or significant recurrent hemoptysis is bronchial artery embolization, if the site of bleeding can be identified. (48)

Pneumothorax

Pneumothoraces occur secondary to air trapping. High alveolar pressure forces air into the lower pressure interstitial spaces, leading to air leak into the pleural space, resulting in symptoms of acute chest pain and dyspnea. (16) The prevalence of having at least one pneumothorax is 3.4% among individuals with CF. (50) Pneumothoraces are more likely in adults and those with advanced lung disease. (50) The initial diagnostic test is chest radiography, but computer tomography may be required to define the extent of a pneumothorax in severe lung disease. (16) Small pneumothoraces may be observed or treated with needle aspiration, but large pneumothoraces require chest tube placement and hospitalization. (48) Pleurodesis is an option for recurrent pneumothoraces, but may complicate later lung transplantation. Following development of a pneumothorax, individuals should refrain from devices that use positive pressure (including vest therapy) as well as pulmonary function testing for at least two weeks, as these can hinder the resolution of the pneumothorax or lead to recurrence. (48)

Chronic Rhinosinusitis and Nasal Polyposis

The epithelium of the sinuses also possess defective CFTR protein, so chronic pan-sinus disease is almost universal among individuals with CF. (51) In contrast, the prevalence of nasal polyposis is more variable and increases with age. (52) Both rhinosinusitis and nasal polyposis result from mucous obstruction of the sinus ostia. Clinical presentations may include chronic headache and facial pressure. Long-term findings may include broadening of the nasal bridge and septal deformation from chronic nasal obstruction. (16) Medical management includes saline nasal irrigation to aid with mucus clearance and nasal steroids to decrease inflammation. Surgical treatment for severe and recurrent disease can improve mucus clearance, but may not necessarily improve lung function. (53)(54)

Distal Intestinal Obstruction Syndrome

Distal Intestinal Obstruction Syndrome (DIOS) is a common gastrointestinal complication in CF. It presents as partial or complete small bowel obstruction secondary to viscous fecal impaction in the distal intestine. Clinical manifestations include abdominal pain and distension, emesis, and a history of decreased stooling. Pathophysiology may be related to CFTR-dependent bile acid secretion and uptake within the distal ileum. (55) Known risk factors for DIOS include dehydration, dietary changes, suboptimal fat absorption (ie inadequate PERT dosing), immobilization, bacterial overgrowth, a prior episode of DIOS, and constipating medications. (55)(56) Differential diagnosis for DIOS includes intussusception, constipation, intestinal adhesions from prior abdominal surgery, volvulus, inflammatory bowel disease and appendicitis. (16)(56) Abdominal radiography, history, and physical examination are usually sufficient to make the diagnosis, but other causes of bowel obstruction should be considered. Management typically includes rehydration and osmotic laxative therapy. More severe cases may require inpatient admission, intravenous fluids, complete bowel rest, and the use of large volumes of polyethylene glycol. Near complete obstruction may require sodium meglumine diatrizoate (Gastrografin) enemas with retrograde lavage and visualization of the terminal ileum by an experienced radiologist. With early diagnosis and implementation of appropriate medical management, surgical interventions are generally not required for DIOS. (16)

Cystic Fibrosis-Related Diabetes Mellitus

Cystic fibrosis leads to disruption of both the exocrine and endocrine functions of the pancreas. Cystic Fibrosis-Related Diabetes Mellitus (CFRD) results from ongoing obstructive damage to the pancreas from thick secretions, which in turn results in fatty infiltration of the pancreas and islet cell destruction. (16) CFRD typically presents after the first decade, and occurs in up to 20% of adolescents and at least 50% of adults with CF. (57) CFRD is typically diagnosed in patients with pancreatic insufficiency and has been associated with increased morbidity and mortality through worse nutritional status and decreased lung function. (58)(59). CFRD is often asymptomatic, and unexplained decreases in growth, weight, or lung function may be related to occult CFRD. Annual screening conducted via 2-hour oral glucose tolerance test should be initiated at age 10 years. (60) Studies have found that fasting levels of hyperglycemia or increased levels of glycated hemoglobin are not sufficiently sensitive for the diagnosis of CFRD, therefore HbA1c levels

should not be used for screening for CFRD. (61) However, HbA1c levels can be used to monitor glucose control in individuals with CFRD. While microvascular complications such as retinopathy, microalbuminuria, and neuropathy may occur with CFRD, similar to other forms of diabetes, ketoacidosis is uncommon. (16)(60) The management of CFRD is focused on glycemic control through insulin therapy; oral anti-hyperglycemic agents are not as effective as insulin in improving long-term outcomes. Nutritional management remains focused on maintaining a high calorie diet while attempting to limit intake of processed carbohydrates to avoid hyperglycemia.

Cystic Fibrosis Liver Disease

Liver disease accounts for 3.4% of overall CF mortality. (2) Approximately 3% of individuals develop CF-related cirrhosis (primarily individuals with pancreatic insufficiency), with a median age of diagnosis of 10 years. (62)(63) Clinical manifestations include cholestasis, cholelithiasis, cirrhosis, portal hypertension, and in severe cases, end-stage liver disease. The pathophysiology is thought to be related to the role of CFTR in promoting bile flow, with abnormal flow leading to cholestasis and biliary fibrosis. (64) Males as well as carriers of Alpha-1 antitrypsin Z allele are at increased risk of developing advanced liver disease. (64)(65) Periodic screening and evaluation for Cystic Fibrosis Liver Disease (CFLD) is critical, as many individuals remain asymptomatic even with advanced cirrhosis. Annual screening of individuals should include assessment of liver function (AST, ALT, and GGT), an ultrasound evaluation and/or evaluation by a gastroenterologist for other causes of liver disease, as indicated. The presence of CFLD should be considered with at least 2 of the following: (1) abnormal physical examination findings such as hepatosplenomegaly, (2) abnormalities of liver function test results above normal reference ranges on at least 3 consecutive determinations during a 12-month period, (3) ultrasound with Doppler evidence of abnormal liver echotexture or portal hypertension, or (4) tissue biopsy has ruled out other causes of liver disease. (66) While there are currently no proven therapies to prevent development or progression of CF-related cirrhosis, ursodiol is frequently used in the management of hyperbilirubinemia in CFLD.

Bone Disease

Another complication of CF is cystic fibrosis-related bone disease, which manifests as low bone density and increased rates of fractures. Poor bone health is likely a result of a combination of factors including malabsorption of fat-soluble vitamins such as vitamin D and K, in addition to poor nutritional status and chronic lung inflammation. Prevention includes optimizing nutritional status as well as encouraging weight-bearing exercise, while dual X-ray absorptiometry is used for screening at-risk patients. Treatment includes aggressive management of potential pulmonary or endocrine co-morbidities, and may include bisphosphonates for those with severe osteopenia. (67)

Depression/Anxiety

Living with a chronic illness, such as CF, can be both challenging and isolating, placing individuals with CF at higher risk for mental health issues. Approximately 15% of all individuals with CF report having either an anxiety disorder or depression, and 41% of individuals with CF report having both conditions. (2) It is recommended that all children

with CF who are 7-11 years old be clinically assessed for depression and anxiety when a caregiver reports clinically elevated symptoms of depression or anxiety, or when there is significant concern for the child exhibiting symptoms of depression or anxiety. Annual screening for depression and anxiety in individuals with CF should begin at 12 years old using the PHQ-9 and GAD-7, respectively. Annual screening is also recommended for caregivers of children with CF. (68) Early identification of mental health difficulties is critical to helping ensure individuals receive referrals to appropriate mental health services in order to receive treatment and maintain their overall health and quality of life.

Preventative Care

Children with CF should receive routine well-child care according to the American Academy of Pediatrics guidelines, including all vaccinations. Annual influenza vaccination is recommended for children ≥ 6 months, as well as for all household members. The use of palivizumab should be considered in all children with CF under 2 years old as prophylaxis against respiratory syncytial virus. (20) Providers should encourage a smoke-free environment for all children with CF and caregivers should be informed of the health effects associated with second-hand smoke exposure.

Transition from Pediatric to Adult CF Care Centers

As most individuals with CF are living into adulthood, the topic of transition from pediatric care centers to adult care centers will continue to be important for patients and their families. One step identified to improve the transition process includes introducing the ideas of self-care skills and transition to adult care during the teenage years. Readiness assessments for self-management skills, including the Transition Readiness Assessment Questionnaire (TRAQ) (69) and transition toolsets have been used to assess patients' readiness to transition to more independent management of their disease. These discussions should also include educational/vocational plans, behavioral risk counseling, screening for depression/anxiety and reproductive health and family planning. Finally, implementation of formal transition-focused visits may be helpful to introduce patients to their new care team in a familiar clinic setting. (70)

Lung Transplantation

Pulmonary disease continues to account for almost 60% of CF-related mortality. (2) Lung transplantation is a surgical option that can extend and improve the quality of life of individuals with CF, but involves extensive evaluation prior to transplant as well as adherence with therapies and lifestyle recommendations to optimize the success of transplanted lungs. Lung transplant confers a survival benefit, (71) with recent reports indicating that individuals with CF are experiencing 9.5 years median survival following lung transplantation. (72) CF providers are recommended to discuss disease trajectory and treatment options including risk and benefits of lung transplantation with individuals with advanced lung disease, and referral to lung transplant centers should be discussed with individuals whose $FEV_1 < 50\%$ predicted or is rapidly declining ($>20\%$ decline in FEV_1 over 12 months). (73) Overall, 6.3% of CF transplants performed in 2018 were among individuals younger than age 18, (2) and criteria for referral are similar to those for adults over 18 years of age. (73) Discussion about lung transplantation may be viewed as another transition by

individuals with CF, and should be facilitated by education, communication and support for the individual and their family. (73)

Conclusion

Individuals affected by CF are living longer and healthier lives, and survival is expected to continue to improve with earlier diagnosis through routine newborn screening, promulgation of evidence-based guidelines, interdisciplinary care centers, and the use of mutation-specific modulator therapies. The primary goals of treatment remain optimization of pulmonary function and nutritional status, and incremental advances in these therapies have had a profound effect on health and quality of life for individuals with CF. Building partnerships with individuals and their families requires recognition of the emotional, social and financial effects of this lifelong disease and effective communication and coordination among primary care physicians and CF care center teams.

Abbreviations:

ACT	airway clearance therapy
ABPA	allergic bronchopulmonary aspergillosis
CF	cystic fibrosis
CFF	Cystic Fibrosis Foundation
CFLD	cystic fibrosis liver disease
CFRD	cystic fibrosis-related diabetes mellitus
CFTR	cystic fibrosis transmembrane conductance regulator
DIOS	distal intestinal obstruction syndrome
IRT	immunoreactive trypsinogen
NBS	newborn screening
PERT	pancreatic enzyme replacement therapy
PE_x	pulmonary exacerbations

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Practice Gap

1. In light of the significant increase in the median age of survival for individuals with cystic fibrosis (CF), both pediatric and adult providers should be familiar with the current recommendations related to optimization of lung health and nutritional status.
2. Providers should also be aware of the development of modulator therapies that target the different basic genetic defects of the disease, which have allowed for personalized therapies that promise continued improvement in outcomes. As of 2019, approximately 90% of individuals with CF have mutations that would benefit from modulator therapy.
3. Clinicians should be familiar with the clinical presentation, diagnosis, and current management of CF as well as the more common disease-related morbidities.

Objectives

After completing this article, readers should be able to:

1. Describe the common clinical manifestations of CF, as well as the laboratory and genetic studies needed to diagnose CF.
2. Recognize the most common presentations of cystic fibrosis-related morbidities.
3. Describe the current recommendations for long-term maintenance of optimal lung health, nutritional status, and other involved organ systems in children with CF.

Summary

- On the basis of consensus, (22) the diagnosis of cystic fibrosis (CF) is based on (1) a positive newborn screening; (2) clinical features consistent with CF (the presence of one or more characteristic phenotypic features of chronic, recurrent sinopulmonary disease, nutritional and gastrointestinal abnormalities, male urogenital abnormalities (e.g., absence of the vas deferens), and salt depletion syndromes, or (3) a positive family history of CF, *and* laboratory demonstrated evidence of CFTR dysfunction, such as elevation of sweat chloride concentration (≥ 60 mmol/L)
- On the basis of consensus, (50) annual oral glucose tolerance tests are recommended for people with CF older than 9 years to screen for CF-related diabetes mellitus.
- On the basis of research evidence, (25) the long-term therapies to maintain optimal lung health for children and adults with CF include control of chronic airways infection and inflammation, clearance of mucous secretions, and, where clinically applicable, treatments aimed at the basic CF genetic defect.

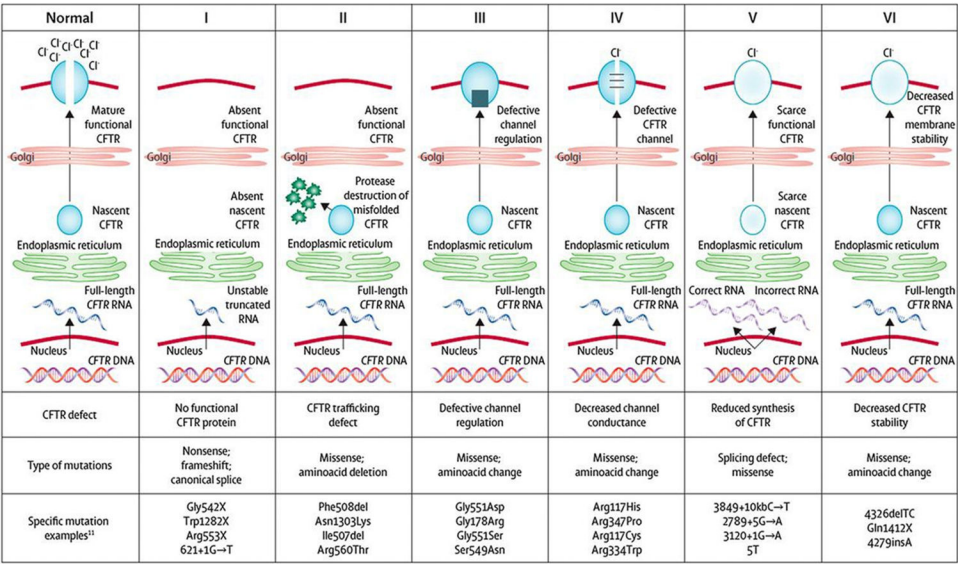


Figure 1. Cystic fibrosis transmembrane conductance regulator (CFTR) mutation classes. CFTR mutations have been grouped into 6 distinct classes based on abnormalities of CFTR synthesis, structure, and function. Reprinted with from: Boyle MP, DeBoeck K. A new era in the treatment of cystic fibrosis: correction of the underlying CFTR defect. *Lancet Respir Med.* 2013;1(2):158–163.

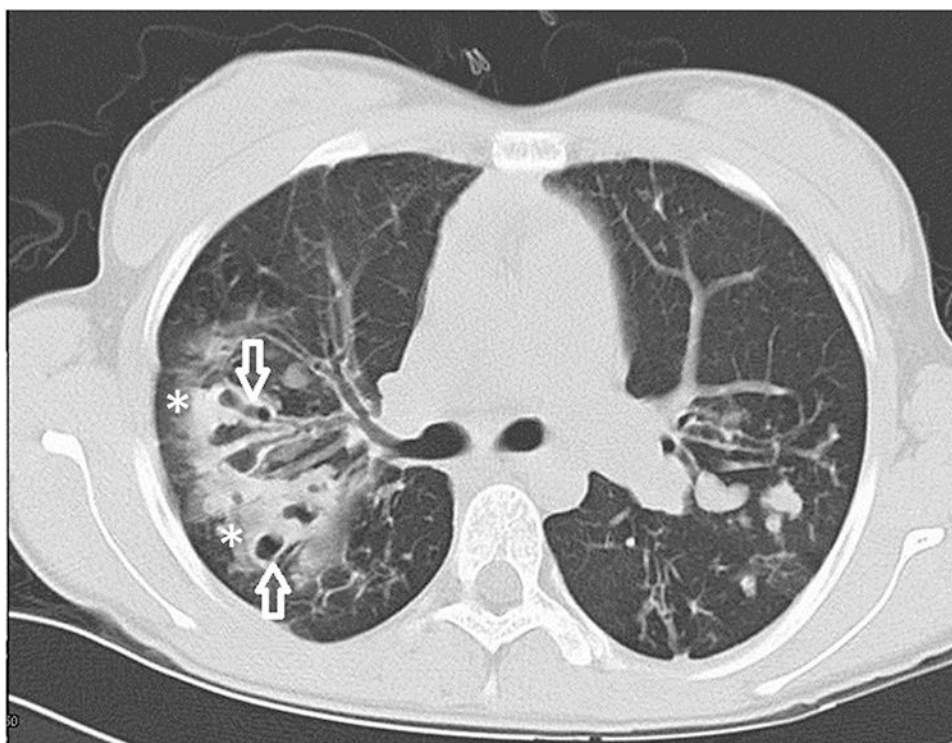


Figure 2. Chest computed tomographic scan of an adolescent girl with cystic fibrosis demonstrates significant bronchiectasis (white open arrows) and mucous plugging (white asterisks). Reprinted from: Paranjape SM, Mogayzel PJ. Cystic fibrosis. *Pediatr Rev.* 2014;35(5).

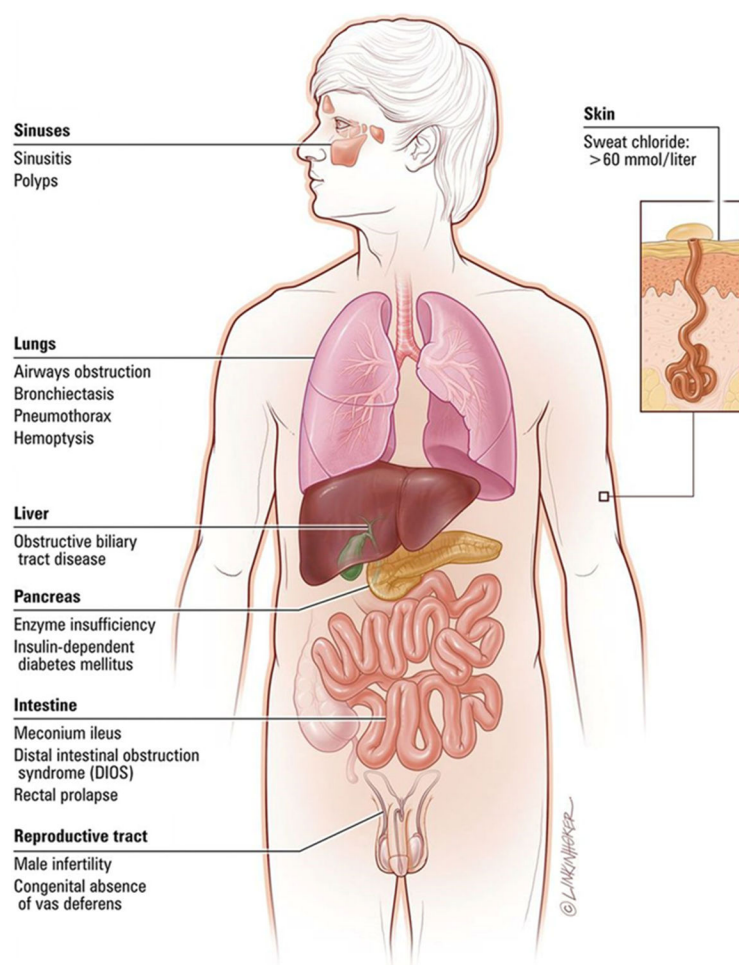


Figure 3.
Common clinical manifestations of cystic fibrosis. Reprinted with permission from Link Studio LLC.

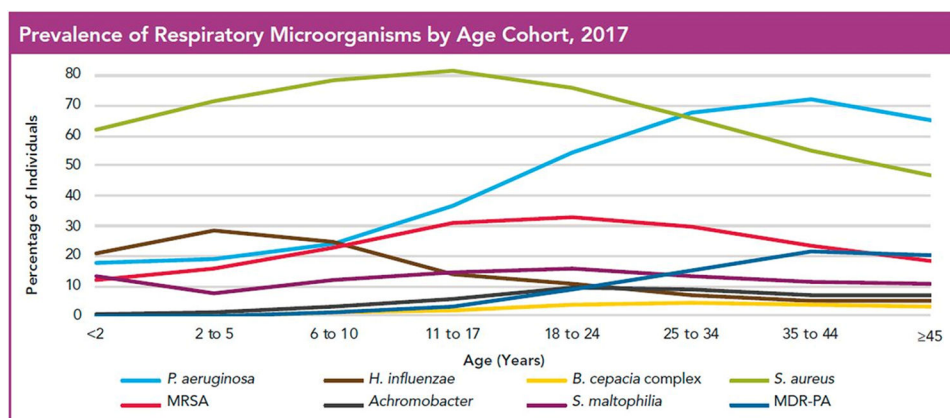


Figure 4. Prevalence of bacterial pathogens in cystic fibrosis. MDR-PA=multidrug-resistant *Pseudomonas aeruginosa*, MRSA=methicillin-resistant *Staphylococcus aureus*. Reprinted from: Cystic Fibrosis Foundation Patient Registry © 2017 Cystic Fibrosis Foundation.